CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- DriverPass wants a system that makes it easier for students to get ready for their driving tests. They're trying to move everything into one place so people can study, take practice tests, and set up driving lessons without dealing with scattered tools or manual scheduling. The idea is to give students a simple online setup where they can work through course material and book time with instructors.
- The system also needs to support the day-to-day work of the staff. The owner, IT admin, and secretary should be able to manage accounts, handle appointments, adjust training packages, and pull reports when they need them. Essentially, the project is meant to create a secure cloud solution that helps students and staff do what they need without extra hassle.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

DriverPass exists because a lot of people struggle with DMV tests. Most of the time, it's not because they can't drive, but because they didn't get enough structured practice or guidance. DriverPass tries to fix that by offering a mix of online learning and real driving instruction.

To support that approach, the system needs a few major pieces:

- Online lessons and practice tests
- Driving sessions with instructors
- Packages that combine digital learning, instructor time, and driving hours

To make all of that work smoothly, the platform has to handle things like:

- Collecting customer info and payments
- Scheduling and tracking driving lessons
- Showing student progress through lessons and tests
- Giving staff different levels of access
- Getting updates from the DMV when rules or test questions change

DriverPass wants everything hosted in the cloud so they don't have to deal with local servers, backups, or security on their own. It also makes the system easier to access from anywhere.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

The finished system should make it clear that the requirements were met. For students, that means being able to:

- Create accounts, sign in, reset passwords, and manage their own lesson appointments
- Use online classes and practice tests, with progress details like test name, time spent, score, and completion status

Staff should be able to:

- Book, update, or cancel driving lessons for students
- Record which instructor, vehicle, date, and time are tied to each session
- Turn training packages on or off
- See activity logs that show who made changes to reservations
- Receive DMV updates when new rules or questions come out
- Reset passwords, block accounts, and manage user roles

The system should work from any device with internet access, and users should be able to download reports if they need them offline.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system needs to run in a web browser and be hosted in the cloud, since DriverPass doesn't want to deal with local servers or backups.
- Pages should load quickly (around two seconds or less), so customers don't get frustrated when booking lessons or checking their progress.
- Scheduling actions like booking or canceling should update almost instantly.
- The system should be able to handle a couple hundred users at the same time without slowing down.
- Users should be able to download reports to work offline, like Liam mentioned with Excel.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should work on common devices like Windows PCs, Macs, tablets, and phones.
- The backend needs a secure cloud database to store user accounts, lesson schedules, instructor info, and test results.
- The system should be able to connect to DMV data sources so DriverPass can get updates when rules or test questions change.
- Everything should run through a browser with no local installation needed.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system needs to clearly separate user roles (student, instructor, secretary, IT admin, owner), so each person only sees what applies to them.
- Scheduling inputs like dates, times, car numbers, and driver assignments should be checked before saving to avoid conflicts.
- If something goes wrong (like overlapping appointments or invalid data), the system should notify the admin.
- Login info should be case-sensitive.
- Activity logs should accurately show who created, changed, or canceled reservations, so Liam can track responsibility.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Admins should be able to add, remove, or update users without touching any code.

- Training packages should be easy to turn on or off from the admin side.
- The system should keep working even when browsers update or new devices come out.
- The IT officer needs access to system logs, account tools, and settings so he can maintain everything.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users need a secure login with a username and password.
- All data sent between the user and the server should be encrypted.
- The system should use role-based access so people only see what they're supposed to.
- If someone tries to brute-force a login, the account should lock after several failed attempts and alert the admin.
- If a user forgets their password, they should be able to reset it through a secure email process.
- Sensitive data like payment info, lesson history, and personal details should be stored securely.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall let users create accounts, log in, and reset passwords.
- The system shall validate user credentials when logging in.
- The system shall let students schedule, update, or cancel driving lessons online.
- The system shall let the secretary schedule lessons by phone or in person.
- The system shall store lesson details like instructor, car, date, time, and session length.
- The system shall track student progress through online lessons and practice tests.
- The system shall let instructors enter notes and comments after each session.
- The system shall let admins manage user accounts, roles, and passwords.
- The system shall let admins activate or deactivate training packages.
- The system shall keep an activity log showing who made, changed, or canceled reservations.
- The system shall let users download reports for offline use.
- The system shall receive DMV updates and notify admins when new rules or test questions come in.
- The system shall show online test progress, including test name, time taken, score, and status.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface needs to run in a web browser and be cloud-based so DriverPass doesn't have to manage servers.
- The main dashboard should include the areas Liam sketched out:
  - Online test progress
  - Student information fields
  - Driver notes
  - Special needs
  - Driver photo and student photo
- There should be simple forms for student registration, scheduling lessons, and contacting DriverPass.
- Instructors should have a place to enter lesson notes and see their schedules.
- Admins should have tools for managing users, packages, and system settings.
- The interface should work on phones, tablets, and computers.
- Navigation should be straightforward, so each user type can find what they need without digging around.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users have a stable internet connection and a modern browser.
- The DMV provides updates in a format the system can use.
- DriverPass will continue using 10 cars and 10 drivers as described in the interview.
- Customers will provide accurate personal and payment information when registering.
- The IT officer will handle account resets and general system maintenance.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Users can't modify data offline. They can only download reports and view them.
- Adding or removing entire training modules isn't part of the first version; only enabling or disabling packages is supported.
- The system depends on the DMV sending updates, so any delays on their end affect DriverPass.
- Budget and timeline limits mean advanced features will have to wait for future releases.
- The system can't verify whether a customer's pickup/drop-off location is correct.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

